**Evaluating Parameter Selection on RareQ's performance**

We implemented a fully label-free parameter selection framework that does not rely on external annotations. This framework is guided by three internal diagnostic criteria (see Methods section in the paper):

(1) Rare-cluster credibility ($Q$): assesses local connectivity and internal cohesiveness.
(2) Cluster independence ($Qc$): measures cluster-level separability by comparing intra-cluster connectivity to total connectivity.
(3) Cross-parameter consensus ($\mathrm{NMI}_{\mathrm{stability}}$): represents the median pairwise NMI between the clustering obtained at a given parameter setting and those obtained under all other settings in the tested parameter grid, providing a label-free consensus measure across the parameter landscape.

To help users navigate the trade-offs among these diagnostics, we provide a ternary plot (Fig. R1a) summarizing their joint behavior across parameter combinations in three single-cell datasets. Overall, RareQ shows robust performance across a broad parameter range. We further highlight the high-performing region—settings that jointly maximize cohesiveness, independence, and stability.

Importantly, parameters are chosen solely based on these label-free diagnostics. We then report $F_1$/NMI only as a post hoc benchmarking to validate the effectiveness of our proposed strategy. As shown in Fig. R1b, parameter settings within the trade-off region yield better classification of rare cells and major cell populations compared with settings from other regions of the ternary plot.

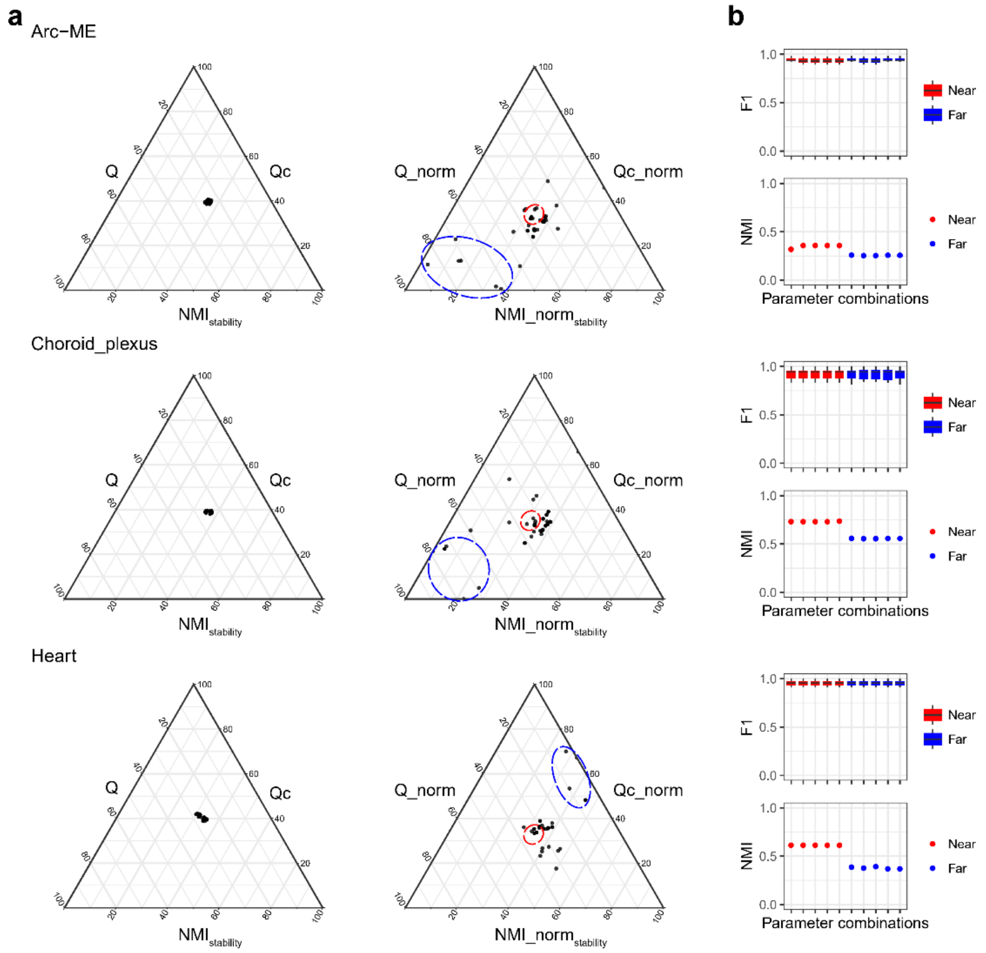


Fig. R1. Label-free parameter selection strategy for RareQ based on ternary plots. (a) The first column presents ternary plots summarizing the relative contributions of three label-free diagnostics ($Q$, $Q_c$, and $NMI_{stability}$) across different parameter combinations. The second column shows zoomed-in ternary plots based on min–max scaling metrics. (b) Boxplots comparing rare-cell detection performance ($F_1$ score) and global clustering accuracy (NMI) between parameter settings located near the center of the ternary plot and those farther from the center.

We also conducted a comprehensive sensitivity analysis by systematically varying *k* parameter from 5 to 30 across 150 simulated and 20 real scRNA-seq datasets.

Our analysis showed that RareQ's performance is highly robust to changes in *k*. In the simulated PBMC datasets, both the $F_1$ scores (measuring rare cell detection accuracy) and NMI values (measuring overall clustering quality) remained stable with only minimal fluctuations (Fig. R2a, b). We observed the same trend in the 20 real datasets, where the $F_1$, precision, recall, and NMI values consistently stayed high across the entire range of *k* values (Fig. R2c, d).

These results demonstrate that RareQ is not overly sensitive to the choice of *k*, and its performance remains stable across a wide range of values.

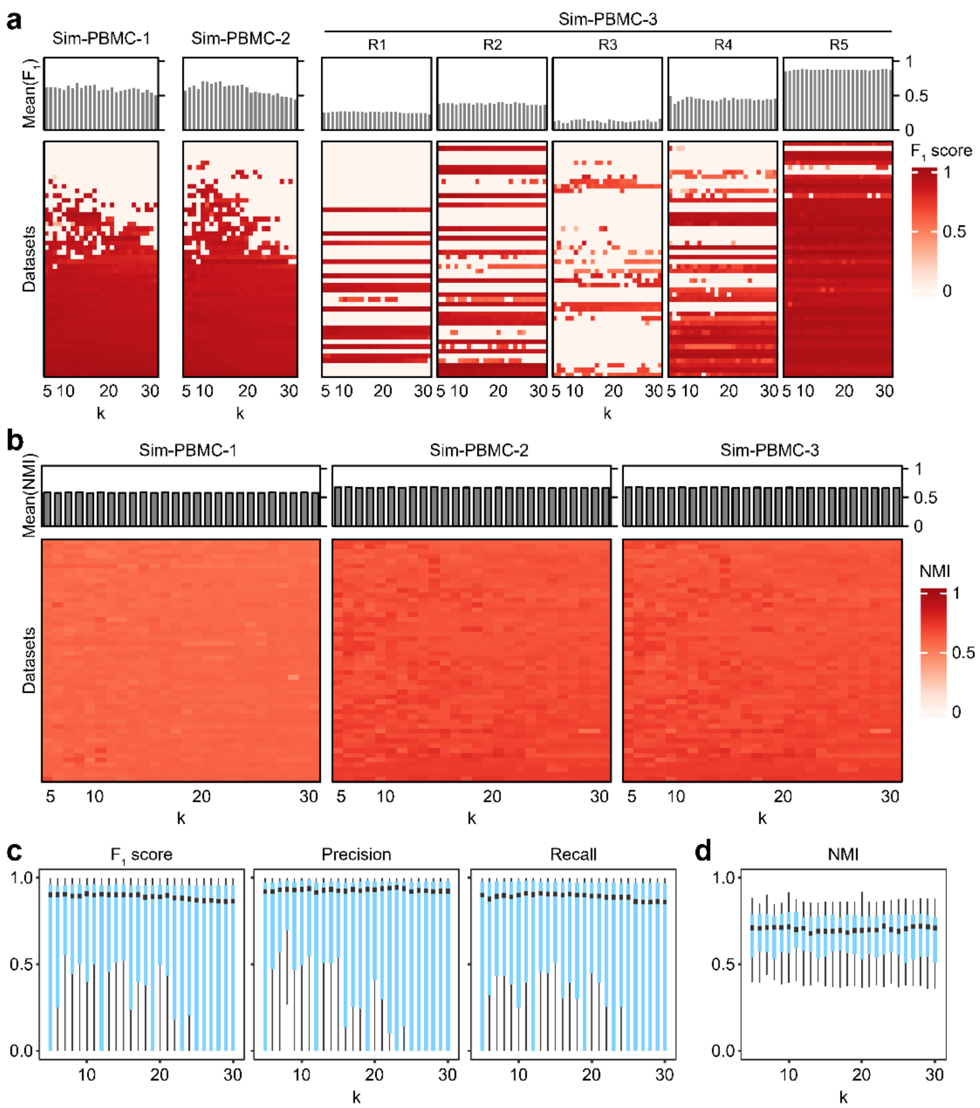


**Fig. R2. Evaluation of RareQ's sensitivity to the choice of *k* value**. **a**. Mean $F_1$ scores (top) and detailed $F_1$ scores (bottom) across different simulation datasets with *k* values ranging from 5 to 30. **b**. NMI values evaluated across the same simulation datasets and *k* ranges as in panel (**a**). **c&d**. Boxplots showing the distribution of $F_1$ score, precision, recall, and NMI at varying *k* values across 20 real scRNA-seq datasets.

Similarly, we performed a comprehensive sensitivity analysis by varying the Q_threshold parameters from 0.1 to 0.9 using both 150 simulated and 20 real single-cell RNA sequencing datasets.

In the three sets of PBMC simulation datasets, $F_1$ scores (reflecting the accuracy of rare cell detection) showed only a slight decline around a threshold of 0.6 (Fig. R3a).

In contrast, NMI values (reflecting overall clustering quality) exhibited a modest increase at 0.4 (Fig. R3b). Across 20 real datasets, $F_1$, precision, and recall remained stable across thresholds, while NMI again showed a small increase at 0.4 but stabilized around 0.6 (Fig. R3c&d).

These findings highlight a clear trade-off: lower *Q* thresholds slightly favor rare cell detection, whereas higher thresholds better preserve overall clustering accuracy.

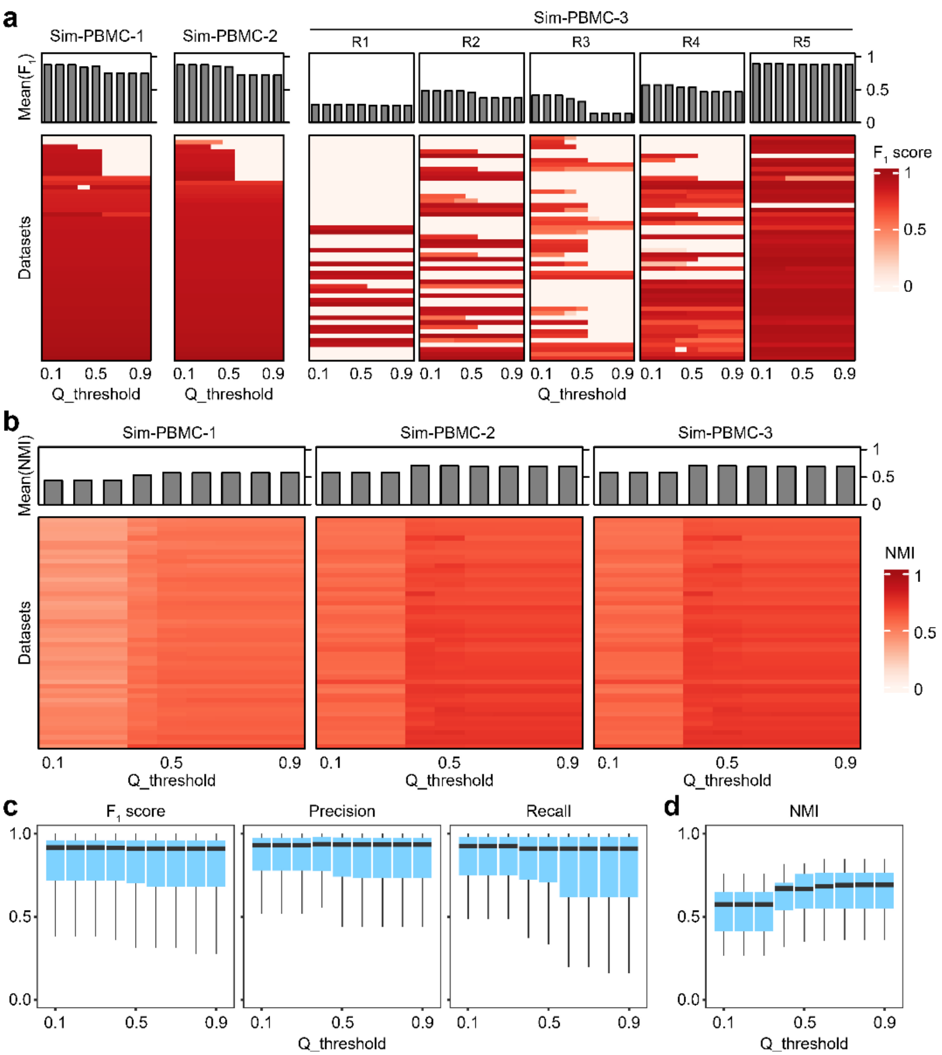


**Fig. R3. Evaluation of RareQ's sensitivity to the choice of Q_threshold value**. **a**. Mean $F_1$ scores (top) and detailed $F_1$ scores (bottom) across different simulation datasets with Q_threshold values ranging from 0.1 to 0.9. **b**. NMI values evaluated across the same simulation datasets and Q_threshold ranges as in panel (**a**). **c&d**. Boxplots showing the distribution of $F_1$ score, precision, recall, and NMI at varying Q_threshold values across 20 real scRNA-seq datasets.

The ratio parameter serves as the threshold for merging clusters. To evaluate its effect, we applied RareQ with values ranging from 0.1 to 0.9 across datasets.

In the PBMC simulation datasets, both $F_1$ scores and NMI remained high and showed minimal sensitivity to changes in this ratio (Fig. R4a, b). In the real scRNA-seq datasets, the average $F_1$ score, precision, and recall were also stable (Fig. R4c, d); however, we observed higher variance at a ratio of 0.1 (Fig. R4c), suggesting that performance may be less reliable under this setting. Taken together, these results indicate that setting the ratio exceeding 0.2 provides stable and reliable performance across most scenarios.

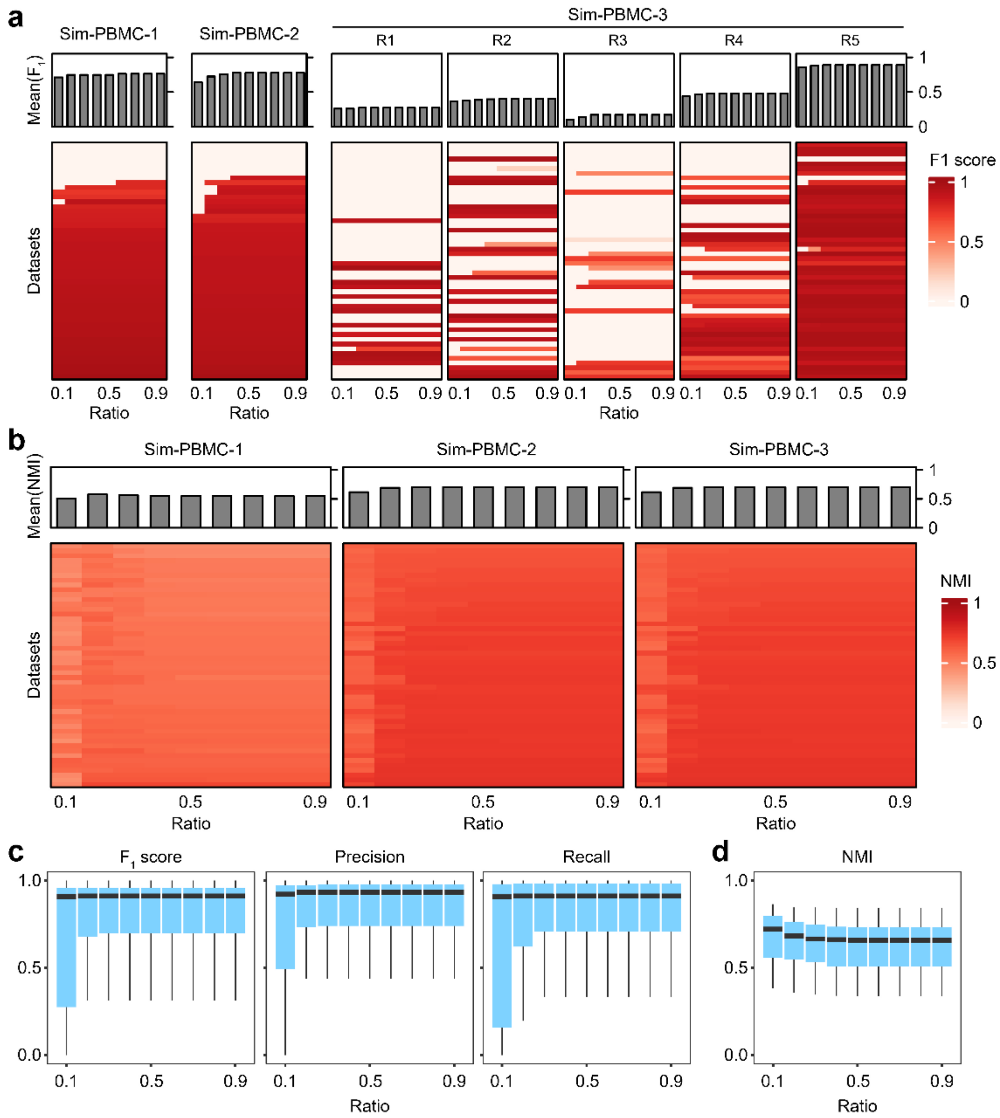


**Fig. R4. Evaluation of RareQ's sensitivity to the choice of ratio value**. **a**. Mean $F_1$ scores (top) and detailed $F_1$ scores (bottom) across different simulation datasets with ratio values ranging from 0.1 to 0.9. **b**. NMI values evaluated across the same simulation datasets and ratio ranges as in panel (**a**). **c&d**. Boxplots showing the distribution of $F_1$ score, precision, recall, and NMI at varying ratio values across 20 real scRNA-seq datasets.